Задание 1

```yaml
apiVersion: batch/v1
kind: CronJob
metadata:
  name: system-info-cron
spec:
  schedule: "*/5 * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
          - name: busybox-container
            image: busybox
            command:
            - sh
            - -c
            - |
              echo "Time: $(date)"
              echo "Host: $(hostname)"
              echo "Process: $(ps aux | wc -l)"
          restartPolicy: OnFailure
```

cronjob.yaml

```
lly@archlinux ~/s/m/t/t/1 (master)> kubectl get cronjob system-info-cron
NAME               SCHEDULE      TIMEZONE   SUSPEND   ACTIVE   LAST SCHEDULE   AGE
system-info-cron   */5 * * * *   <none>     False     1        42s             5m43s
```

kubectl get cronjob system-info-cron

```
lly@archlinux ~/s/m/t/t/1 (master)> kubectl get jobs
NAME                          STATUS     COMPLETIONS   DURATION   AGE
system-info-cron-29591290     Complete   1/1           69s        6m13s
system-info-cron-29591295     Running    0/1           73s        73s
```

kubectl get jobs

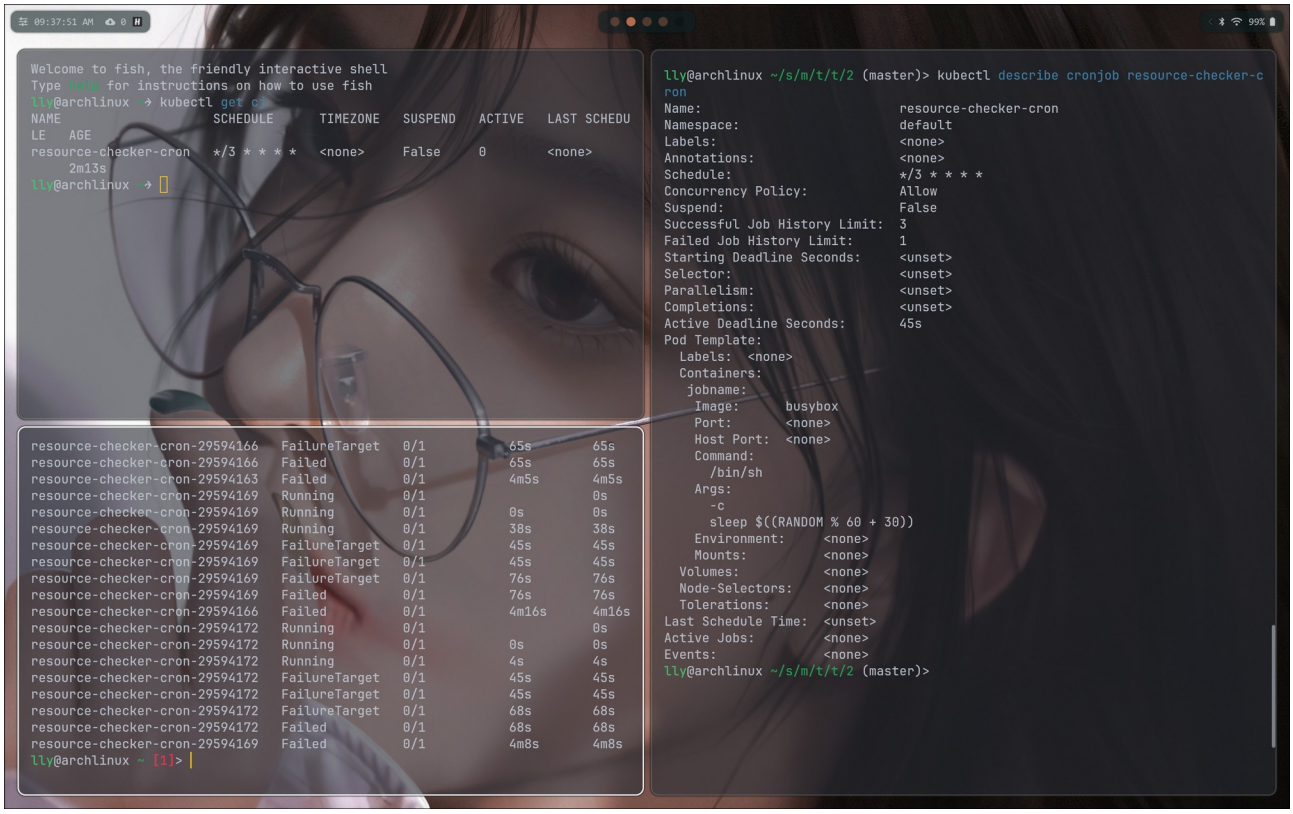
```
lly@archlinux ~/s/m/t/t/1 (master)> kubectl logs -l job-name=system-info-cron-29591290
Time: Mon Apr  6 12:11:07 UTC 2026
Host: system-info-cron-29591290-7ktzz
Process: 5
```

kubecrl logs -l job-name=<job_name>

Задание 2

```yaml
apiVersion: batch/v1
kind: CronJob
metadata:
  name: resource-checker-cron
  namespace: default
spec:
  schedule: "*/3 * * * *"
  jobTemplate:
    spec:
      activeDeadlineSeconds: 45
      backoffLimit: 2
      template:
        spec:
          containers:
          - name: jobname
            image: busybox
            command: ['/bin/sh']
            args: ['-c', 'sleep $((RANDOM % 60 + 30))']
          restartPolicy: OnFailure
```

cronjob.yaml



Задание 3

```yaml
apiVersion: batch/v1
kind: CronJob
metadata:
  name: sequntial-processor-cron
spec:
  schedule: "*/4 * * * *"
  concurrencyPolicy: Forbid
  jobTemplate:
    spec:
      completions: 4
      parallelism: 1
      template:
        spec:
          containers:
          - name: busybox-container
            image: busybox
            command: ["/bin/sh"]
            args: ["-c", "echo Processing item $((RANDOM % 100)); sleep 30"]
          restartPolicy: OnFailure
```

cronjob.yaml

```
lly@archlinux ~/s/m/t/t/3 (master)> kubectl get cj sequntial-processor-cron -o yaml | grep concurrencyPolicy
      {"apiVersion":"batch/v1","kind":"CronJob","metadata":{"annotations":{},"name":"sequntial-processor-cron","namespace":"default"},"spec":{"concurrencyPolicy":"
Forbid","jobTemplate":{"spec":{"completions":4,"parallelism":1,"template":{"spec":{"containers":[{"args":["-c","echo Processing item $((RANDOM % 100)); sleep 30"],
"command":["/bin/sh"],"image":"busybox","name":"busybox-container"}],"restartPolicy":"OnFailure"}}}},"schedule":"*/4 * * * *"}}
  concurrencyPolicy: Forbid
lly@archlinux ~/s/m/t/t/3 (master)> kubectl get pods 0-w
Error from server (NotFound): pods "0-w" not found
lly@archlinux ~/s/m/t/t/3 (master) [1]> kubectl get pods -w
NAME                                      READY   STATUS      RESTARTS   AGE
sequntial-processor-cron-29595384-k4hmp   1/1     Running     0          29s
sequntial-processor-cron-29595384-nlhbk   0/1     Completed   0          100s
sequntial-processor-cron-29595384-z2lqh   0/1     Completed   0          64s
sequntial-processor-cron-29595384-k4hmp   0/1     Completed   0          33s
sequntial-processor-cron-29595384-k4hmp   0/1     Completed   0          34s
sequntial-processor-cron-29595384-5hzx9   0/1     Pending     0          0s
sequntial-processor-cron-29595384-5hzx9   0/1     Pending     0          0s
sequntial-processor-cron-29595384-k4hmp   0/1     Completed   0          35s
sequntial-processor-cron-29595384-5hzx9   0/1     ContainerCreating   0          0s
sequntial-processor-cron-29595384-5hzx9   1/1     Running             0          4s
sequntial-processor-cron-29595384-5hzx9   0/1     Completed           0          34s
sequntial-processor-cron-29595384-5hzx9   0/1     Completed           0          35s
sequntial-processor-cron-29595384-5hzx9   0/1     Completed           0          36s
sequntial-processor-cron-29595388-xms6w   0/1     Pending             0          0s
sequntial-processor-cron-29595388-xms6w   0/1     Pending             0          0s
sequntial-processor-cron-29595388-xms6w   0/1     ContainerCreating   0          0s
sequntial-processor-cron-29595388-xms6w   1/1     Running             0          4s
```

kubectl get cj sequntial-processor-cron -o yaml | grep concurrencyPolicy && kubectl get pods -w

Задание 4

```yaml
apiVersion: batch/v1
kind: CronJob
metadata:
  name: replace-cron
spec:
  schedule: "*/2 * * * *"
  concurrencyPolicy: Replace
  jobTemplate:
    spec:
      template:
        spec:
          containers:
          - name: busybox-container
            image: busybox
            command: ["/bin/sh"]
            args: ["-c", "echo sleeping...; sleep 150"]
          restartPolicy: OnFailure
```

cj-replace.yaml (остальные файлы выглядят идентично, за исключением concurrencyPolicy и name)

```
lly@archlinux ~/s/k/t/t/4 (master)> kubectl get cj -o wide
NAME           SCHEDULE      TIMEZONE   SUSPEND   ACTIVE   LAST SCHEDULE   AGE     CONTAINERS          IMAGES    SELECTOR
allow-cron     */2 * * * *   <none>     False     2        31s             6m31s   busybox-container   busybox   <none>
forbid-cron    */2 * * * *   <none>     False     1        2m31s           6m26s   busybox-container   busybox   <none>
replace-cron   */2 * * * *   <none>     False     1        31s             6m37s   busybox-container   busybox   <none>
```

kubectl get cj -o wide

```
lly@archlinux ~/s/k/t/t/4 (master)> kubectl get jobs --sort-by=.metadata.creationTimestamp
NAME                   STATUS     COMPLETIONS   DURATION   AGE
forbid-cron-29595470   Complete   1/1           5m20s      13m
forbid-cron-29595474   Complete   1/1           3m44s      8m21s
allow-cron-29595476    Complete   1/1           5m1s       7m42s
allow-cron-29595478    Complete   1/1           3m12s      5m42s
forbid-cron-29595478   Complete   1/1           3m20s      4m37s
allow-cron-29595480    Complete   1/1           2m49s      3m42s
allow-cron-29595482    Running    0/1           102s       102s
replace-cron-29595482  Running    0/1           102s       102s
forbid-cron-29595482   Running    0/1           77s        77s
```

kubectl get jobs —sort-by=.metadata.creationTimestamp

Контрольные вопросы

В чем основное отличие обычной Job от CronJob? — Job — простое выполнение задачи, CronJob — периодичное выполнение задачи по заданном графику

Какой формат используется для задания расписания в CronJob? Что означают 5 полей? — используется Cron формат. 5 полей: минута, час, день, месяц, день недели

Какие три политики параллелизма (concurrency policy) существует у CronJob, и в чем их разница? — Allow, Forbid, Replace. Allow — запускает задачу, даже если прошлая не завешила свое выполнение, Forbid — пропускает новую задачу, если прошлая не завершила свое выполнение, Replace — останавливает текущую задачу и запускает новую.

Что произойдет, если выполнение задачи, инциированной CronJob, превысит activeDeadlineSeconds? — Job принудительно завершится с ошибкой

Где настраиваются параметры Parallelism и Completions для задача, создавемых CronJob? — spec.jobTemplate.spec

Как можно посмотреть логи конкретной Job, которая была создана CronJob? — kubectl logs -l job-name=<job_name>

Что произойдет, если CronJob будет запланирована на запуск в момент, когда кластер k8s недоступен? — если запуск не укладывается в activeDeadlineSeconds, то задачи пропустятся

Какие команды вы бы использовали для диагностики проблем с CronJob, которая не создает Jobs? — get jobs, describe